

Lecture 11 – Completeness for exponential families

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11.1 Complete sufficient statistics in exponential families

How can we recognize a complete sufficient statistic without checking every possible function a ? The exponential-family representation gives a useful sufficient condition.

Theorem 1 (Completeness in a one-parameter exponential family). Suppose $X_1, \dots, X_n$ are i.i.d. with density or mass function

$$f_{\theta}(x) = h(x) \exp\{b(\theta)R(x) - c(\theta)\},$$

on a support that does not depend on θ . If the natural-parameter set $\{b(\theta) : \theta \in \Theta\}$ contains a nonempty open interval, then

$$T = \sum_{i=1}^n R(X_i)$$

is sufficient and complete, and therefore minimal sufficient.

Sufficiency follows from factorizing the joint density:

$$f_{\theta}(\underline{x}) = \left(\prod_{i=1}^n h(x_i) \right) \exp \left\{ b(\theta) \sum_{i=1}^n R(x_i) - nc(\theta) \right\}.$$

We state the completeness part without proof. Its key hypothesis is that the natural parameter varies over an interval: the condition $\mathbb{E}_{\theta}[a(T)] = 0$ must hold across a sufficiently rich family of distributions to force $a(T) = 0$.

Example 1 (Finishing the Poisson example). For $X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$, $\theta > 0$,

$$f_{\theta}(x) = \frac{1}{x!} \exp\{x \log \theta - \theta\}, \quad x = 0, 1, 2, \dots$$

Here $R(x) = x$ and $b(\theta) = \log \theta$ ranges over $\mathbb{R}$. Thus $T = \sum_i X_i$ is complete and sufficient. The estimator from Lecture 9,

$$\boxed{\left(1 - \frac{1}{n}\right)^T},$$

is therefore the UMVUE of $e^{-\theta}$ for $n \geq 2$: it is an unbiased, bounded function of a complete sufficient statistic. We have now justified optimality among *all* unbiased estimators, not just improvement over the proportion of zeros.

Example 2 (Bernoulli sampling: correcting the plug-in estimator). Suppose $X_i \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, $0 < \theta < 1$. We have

$$f_\theta(x) = \exp\left\{x \log \frac{\theta}{1-\theta} + \log(1-\theta)\right\}, \quad x \in \{0, 1\}.$$

The natural parameter $b(\theta) = \log(\theta/(1-\theta))$ ranges over $\mathbb{R}$, so $T = \sum_i X_i$ is complete and sufficient. Since $\bar{X}_n = T/n$ is unbiased for θ , it is the UMVUE of θ .

Now suppose $n \geq 2$ and the target is $g(\theta) = \theta(1-\theta)$. The plug-in estimator is $\bar{X}_n(1-\bar{X}_n)$, but

$$\begin{aligned} \mathbb{E}_\theta[\bar{X}_n(1-\bar{X}_n)] &= \mathbb{E}_\theta[\bar{X}_n] - \mathbb{E}_\theta[\bar{X}_n^2] \\ &= \theta - \left(\frac{\theta(1-\theta)}{n} + \theta^2\right) \\ &= \frac{n-1}{n}\theta(1-\theta). \end{aligned}$$

Multiplying by $n/(n-1)$ gives an unbiased function of T . Hence the UMVUE is

$$\boxed{\frac{n}{n-1}\bar{X}_n(1-\bar{X}_n) = \frac{T(n-T)}{n(n-1)}}.$$

No conditioning calculation was needed: once we found an unbiased function of the complete sufficient statistic, Lehmann–Scheffé did the rest.

Example 3 (Normal mean with known variance). Suppose $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$, where $\sigma^2 > 0$ is known. Write

$$f_\theta(x) = \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{\theta}{\sigma^2}x - \frac{\theta^2}{2\sigma^2}\right\}.$$

The natural parameter θ/σ^2 ranges over $\mathbb{R}$, so $\sum_i X_i$ is complete and sufficient. Its one-to-one transformation $\bar{X}_n$ is also complete and sufficient. Thus $\bar{X}_n$ is the UMVUE of θ , while

$$\mathbb{E}_\theta[\bar{X}_n^2] = \theta^2 + \frac{\sigma^2}{n} \implies \boxed{\bar{X}_n^2 - \frac{\sigma^2}{n} \text{ is the UMVUE of } \theta^2}.$$

This reinforces Lecture 8: a nonlinear function of a UMVUE need not be unbiased for the transformed target.

11.2 Completeness and minimal sufficiency

Minimal sufficiency says we cannot compress a sufficient statistic further, except by an equivalent representation, while keeping sufficiency. Completeness says there is at most one unbiased function of the statistic for each target. These are different claims: maximal sufficient compression need not give uniqueness of unbiased answers.

Example 4 (Minimal sufficient, but not complete). Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(\theta, \theta+1)$, where $\theta \in \mathbb{R}$ and $n \geq 2$. We have shown before that $T = (X_{(1)}, X_{(n)})$ is minimal sufficient. But T is not complete. To show this, we use that $\text{Unif}(\theta, \theta+1)$ is a location family. Write $X_i = \theta + Z_i$ with $Z_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$. We know that $h(T) = X_{(n)} - X_{(1)} = Z_{(n)} - Z_{(1)}$ is *ancillary*: its distribution does not depend on θ . Thus for every θ , we have $\mathbb{E}_\theta[X_{(n)} - X_{(1)}] = c$ for some constant c (working out the math gives $c = (n-1)/(n+1)$). Therefore, the nonzero function $a(T) = X_{(n)} - X_{(1)} - c$ has $\mathbb{E}_\theta[a(T)] = 0$ for all θ , so T is not complete.

Theorem 2 (Complete and sufficient implies minimal sufficient). Suppose T is a statistic or a vector of statistics. If T is sufficient and complete, then it is minimal sufficient, using our convention that equalities may ignore events of probability zero.

Proof when T has finite variance. We give the proof assuming that T (or each component, if T is a vector) has finite variance for every θ . Let S be any sufficient statistic. We must show that T is a function of S .

First suppose T is a univariate statistic. Define $g(\theta) = \mathbb{E}_\theta[T]$. Then T is an unbiased function of the complete sufficient statistic T , so Lehmann–Scheffé implies that T is the UMVUE of $g(\theta)$.

Since S is sufficient, Rao–Blackwell gives another unbiased estimator $\mathbb{E}[T | S]$ with

$$\text{Var}_\theta(\mathbb{E}[T | S]) \leq \text{Var}_\theta(T).$$

But T is the UMVUE, so equality holds, and total variance gives

$$0 = \text{Var}_\theta(T) - \text{Var}_\theta(\mathbb{E}[T | S]) = \mathbb{E}_\theta[\text{Var}_\theta(T | S)].$$

It follows that $\text{Var}_\theta(T | S) = \mathbb{E}_\theta[(T - \mathbb{E}[T | S])^2 | S] = 0$ with probability 1 for every θ , which implies $T = \mathbb{E}[T | S]$ with probability 1 for every θ . Sufficiency of S ensures that $\mathbb{E}[T | S]$ does not depend on θ . Thus T is a function of S , as required.

For a vector $T = (T_1, \dots, T_k)$, each T_j is an unbiased function of the complete sufficient statistic T for the target $g_j(\theta) = \mathbb{E}_\theta[T_j]$. Apply the same argument to each coordinate to obtain $T_j = \mathbb{E}[T_j | S]$. Thus the whole vector is a function of S . $\square$

We therefore have

$$\boxed{\text{complete and sufficient} \implies \text{minimal sufficient}},$$

but the converse fails, as Example 4 shows. A complete sufficient statistic gives both the coarsest sufficient representation and uniqueness of unbiased functions.

Proposition 1 (Functions preserve completeness). If T is complete and $S = r(T)$, then S is complete.

Proof. If $\mathbb{E}_\theta[a(S)] = 0$ for every θ , then $\mathbb{E}_\theta[a(r(T))]$ = 0 for every θ . Apply completeness of T to the function $t \mapsto a(r(t))$ to obtain $a(S) = a(r(T)) = 0$ with probability 1 for every θ . $\square$

Coarsening preserves completeness, but it need not preserve sufficiency. For example, a constant statistic is complete, but generally not sufficient. A one-to-one transformation preserves both properties, since it retains exactly the same information.